

Activated carbon derived from coconut coir pith as high performance supercapacitor electrode material

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ABSTRACT

In this study, activated carbons were synthesized from coconut coir pith, an inexpensive and abundant type of biomass in Thailand. The coconut coir pith was activated by a combined physical and chemical activation route, using NaOH-to-char ratios of 1:1, 2:1 and 3:1 by weight. A high surface area of $2056 \text{ m}^2 \text{ g}^{-1}$ was achieved with NaOH-to-char weight ratio of 3:1. Using this activated carbon in a symmetrical cell with aqueous electrolyte, high capacitance values were achieved. The specific capacitance values obtained were 232.3 F g^{-1} at 0.1 A g^{-1} and 192.0 F g^{-1} at 1 A g^{-1} . Long-term cycling tests at 0.5 A g^{-1} showed 95.4% capacitance retention after 5000 cycles. The high specific capacitance was attributed to the high micropore volume. The superior rate capability was from the interconnected pore structure of the activated carbon which allowed for efficient ion transport.

1. Introduction

Supercapacitors are power devices that can be charged and discharged within seconds or minutes. Interest in supercapacitors has increased significantly in recent years due to its potential for utilization in a wide range of applications. They can complement or replace batteries in electric vehicles, energy storage, uninterruptible power supply or load-leveling [1]. Traditionally, supercapacitors consist of carbon-based electrodes, an aqueous or organic electrolyte and an ion-permeable separator. High-surface area carbons, such as activated carbons, carbon fibers and carbon aerogels are used as electrode materials in supercapacitors [2]. Energy is stored in carbon-based electrodes by the accumulation of opposite charges on either side of the electrode/electrolyte interface, separated by an electric double layer [3].

A large number of studies have been carried out to explore the use of various types of biomass materials for use as supercapacitor electrode material as alternatives to coconut shell-derived carbon. Kalyani and Anitha [4] have given a thorough review of the literature on this topic. These tend to be solids that are considered agricultural waste, e.g., pits, seeds, hulls/shells and used residue, or materials that are available in large quantities and may not be fully utilized in other applications. Biomass precursors continue to be widely studied for its use as a source of carbon for supercapacitors as they are inexpensive, available in large quantities, renewable and have a naturally porous structure [5].

Thailand has an agriculture-based economy, producing nearly 60 million tons of agricultural waste every year. Coconut waste

accounted for approximately 1.2 million tons in 2010 [6]. Coconut coir pith (CCP) accounts for 70% of the husk and is separated as waste in the coconut coir fiber separation process [7]. Typically, CCP is used as a gardening material, but is interesting as a precursor for carbon supercapacitor electrodes due to its high carbon content and relatively low ash content. The cellulose, hemicellulose and lignin contents are 15.70–36.00, 5.40–14.50 and 38.00–59.50 wt%, respectively [8–12]. The high lignin content provides CCP an advantage when producing activated carbon, because lignin has a carbon content of nearly 60 wt% [13,14]. Lignin and cellulose also contribute to the structural stability of the material, which may translate into a more stable supercapacitor electrode structure [10]. CCP is also inexpensive, \$0.15–0.20/kg compared to \$0.25/kg for coconut shells [15].

Only a few studies have been done on activated carbons made from coconut coir pith and are typically for dye adsorption or air and water purification purposes [9,12,16]. In this study, we examined the properties of CCP-derived activated carbon as an electrode material in supercapacitors. Using combined physical and chemical activation, the effects of amount of activating agent on the physical and electrochemical performance of the CCP-derived activated carbons were investigated.

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2. Materials and methods

2.1. Preparation of biomass precursor

CCP was obtained from a local market (Talat Thai) in Pathum Thani province in Thailand. The as-received CCP was rinsed with tap water to remove soil and other natural contaminants, ultrasonicated in DI water then dried at 110 °C for 16 h. The dried CCP was ground with a disc grinding mill (Herzog HSM100A) for 2 min and sequentially sieved through 30 (600 μm) and 50 (300 μm) wire mesh.

2.2. Carbonization procedure

Approximately 9 g of ground CCP was placed in a 50 ml alumina combustion boat. The ground CCP was carbonized in an alumina reactor at 650 °C for 2 h (ramp rate 10 °C min⁻¹) in N₂ gas (150 ml min⁻¹ flow rate) then allowed to cool to room temperature in N₂ atmosphere.

2.3. Activation procedure

Approximately 1.0 g of CCP char was mixed with NaOH (99%, VWR chemicals) with a weight ratio of NaOH:CCP char = 1:1, 2:1 or 3:1. The mixture was transferred to an alumina crucible, 15 ml of DI water was added, then the mixture was stirred at 200 °C for 2 h. The mixture was then allowed to dry at 150 °C for 3 h. The dried mixture was ground again with mortar and pestle then transferred to an alumina combustion boat. In an alumina tube reactor, the sample was heated in N₂ gas (50 ml min⁻¹) at 5 °C min⁻¹ to an activation temperature of 700 °C. The gas was then switched to CO₂ (40 ml min⁻¹), held for 1 h, then switched back to N₂ (50 ml min⁻¹) and the furnace was ramped down to room temperature (10 °C min⁻¹). After activation, the carbon sample was washed using 0.5 M HCl solution, 1 M NaOH solution and DI water consecutively until a neutral pH was reached. The sample was then dried at 110 °C for 16 h and ground with mortar and pestle. The carbon was rinsed with acidic and alkaline solutions in order to remove metallic and other impurities such as Si, Ca and Mg compounds. The activated carbons made from NaOH:CCP char weight ratios of 1:1, 2:1 and 3:1 were denoted as AC-1, AC-2 and AC-3, respectively.

2.4. Physical and chemical characterization

Raw CCP was studied using thermogravimetric analysis (TGA; NETZSCH STA 449C thermal analyzer), proximate analysis (Van Soest method), elemental (CHNO) analysis (LECO CHN-628 elemental analyzer and LECO TruSpec Micro analyzer) and scanning electron microscopy (SEM; Hitachi SU8030). The surface area and pore size distribution of activated samples were studied using the Brunauer–Emmett–Teller (BET) technique (Micromeritics ASAP 2460). The total pore volume (V_{tot}) was determined at $P/P_0 = 0.95$. The micropore volume (V_{micro}) and pore size distribution were determined by the Horvath–Kawazoe (H–K) equation and non-local density functional theory (NLDFT), respectively. The activated samples were additionally characterized by Raman spectroscopy with 532 nm laser wavelength (Bruker Optics Senterra Dispersive Raman Microscope), scanning electron microscopy (SEM; Hitachi SU8030), transmission electron microscopy (TEM; JEOL JEM-2010), Fourier transform infrared spectroscopy (FT-IR, Perkin Elmer Spectrum One FT-IR spectrometer), X-ray diffraction (XRD; PANalytical X'Pert Pro 2 kW diffractometer with Cu K α radiation), X-ray photoelectron spectroscopy (XPS; Kratos Axis Ultra DLD with Al K α radiation) and inductively-coupled plasma with mass spectroscopy (ICP-MS; Perkin-Elmer NexION 350X).

2.5. Electrochemical characterization

In order to study their performance as electrode material for electrochemical capacitors, the activated carbons from CCP were mixed

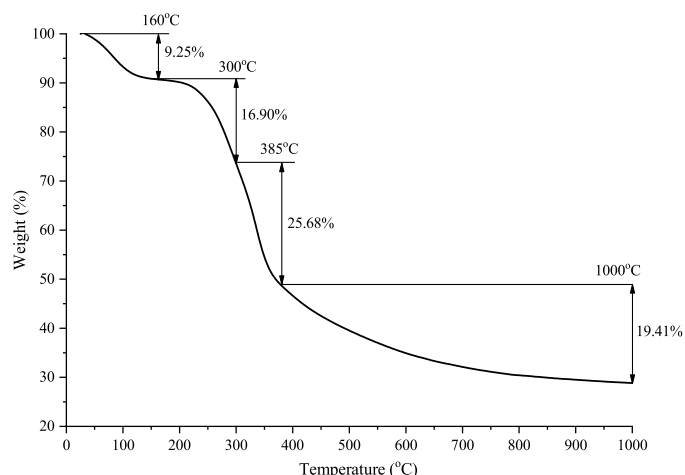


Fig. 1. Thermogravimetric analysis curve of coconut coir pith from room temperature to 1000 °C at a rate of 10 °C min⁻¹ under N₂ flow (20 ml min⁻¹).

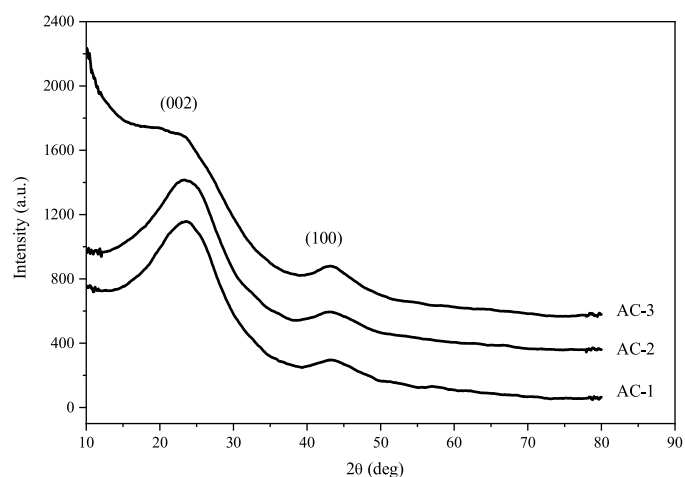


Fig. 2. X-ray diffraction patterns of activated carbon samples AC-1, -2 and -3.

Table 1

BET surface area, total pore volume and micropore volume of AC-1, -2 and -3.

Sample	BET surface area (m ² g ⁻¹)	V_{total} (cm ³ g ⁻¹)	$V_{\text{micropore}}$ Volume (cm ³ g ⁻¹)	% by volume
AC-1	633.2	0.39	0.33	84.6
AC-2	860.1	0.51	0.44	86.3
AC-3	2056.0	1.17	1.06	90.6

with conductive carbon (Super C65, Timcal) and binder (polyvinylidene fluoride, Arkema) in an 80:10:10 weight ratio. For charge–discharge, cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS) measurements, electrodes were formed into 11 mm diameter pellets using a uniaxial hydraulic press (pressure $\approx$ 150 MPa). Each pellet contained 20 mg of powder and was approximately 200 μm in thickness. Symmetrical cells were assembled using Swagelok cells with a nonwoven glass fiber separator (Whatman GF/F, Sigma-Aldrich) as separator and 1 M H₂SO₄ solution as the electrolyte. For long-term cycling tests, pouch cells were used. Slurries were formed from the powder mixtures using *N*-methyl-2-pyrrolidone as solvent. Electrodes were casted onto titanium foil (0.1 mm thick). Each electrode was 2.0 cm in length and width, approximately 40–60 μm in thickness, with an active material loading of 15–20 mg. CV measurements were performed between 0.0 and 1.0 V using potential scan rates between 5–100 mV s⁻¹ and a step size of 0.5 mV. EIS measurements were

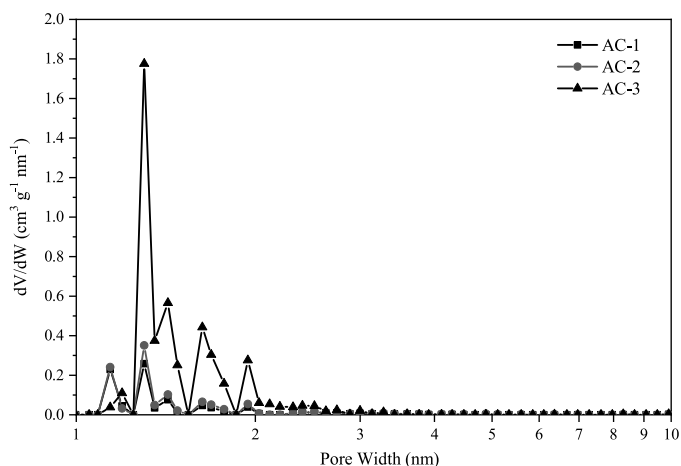


Fig. 3. Micropore size distribution of activated carbon samples AC-1, -2 and -3.

carried out in the frequency range of 10^{-2} – 10^5 Hz and a signal amplitude of 10 mV. CV and EIS measurements were performed using an Ecochemie Autolab PGSTAT302 potentiostat/galvanostat. Galvanostatic charge–discharge curves were obtained using currents in the range of 0.1–10 A g $^{-1}$ with cut-off potentials of 0.0 and 1.0 V (Maccor 4000 battery analyzer). The specific capacitance values (C_{sp} , F g $^{-1}$), defined as the capacitance per unit mass for one electrode, were calculated from galvanostatic charge–discharge curves using Eq. (1) [2,17].

$$C_{sp} = \frac{4I\Delta t}{\Delta Vm} \quad (1)$$

where I (A) is the discharge current, Δt (s) is the discharge time, ΔV (V) is the potential drop during discharge (with the IR drop subtracted) and m (g) is the mass of the active material in both electrodes.

3. Results and discussion

3.1. Raw material characterization

To better understand the structure of the raw material and determine appropriate conditions for carbonization and activation, thermal analysis was performed (Fig. 1) by heating raw CCP from room temperature to 1100 °C at a rate of 10 °C min $^{-1}$ under N $_2$ flow of 20 ml min $^{-1}$. Four major decomposition stages were found: at < 160 °C, water is desorbed and removed from the structure; between 160 °C and 300 °C, cellulose structural water is expelled, hemicellulose decomposes and cellulose begins to decompose; between 300 °C and 385 °C, which is where the largest mass loss occurs, cellulose continues to decompose, lignin begins decomposition, as well as chain scissions or depolymerization and breaking of C–O and C–C bonds with glucopyranose rings take place, resulting in water, CO and CO $_2$ formation. Finally, above 385 °C, lignin continues to decompose up to 500 °C while graphitic layers are formed via carbon aromatization [9,18]. The carbonization temperature chosen for this study was 650 °C in order to ensure complete decomposition of lignin and aid the graphitization of carbon, which was expected to improve its conductivity. The surface area of the carbonized CCP was ≈ 600 m 2 g $^{-1}$.

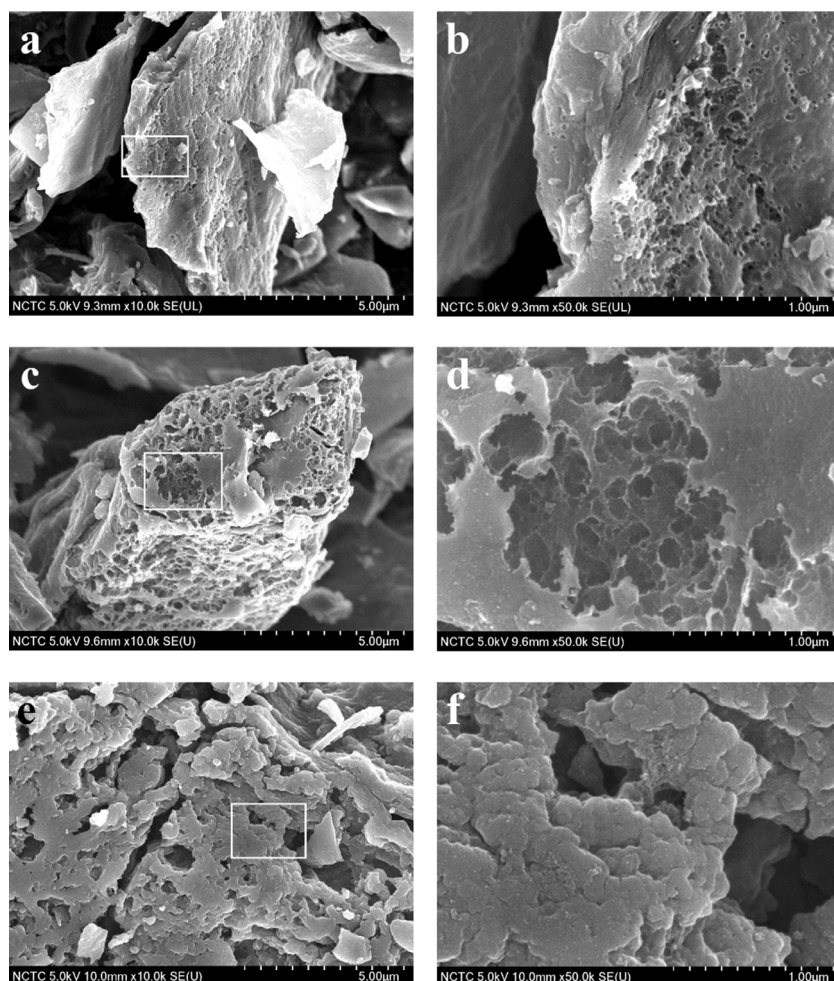


Fig. 4. Scanning electron micrographs of AC-1 (a, b), AC-2 (c, d) and AC-3 (e, f) at different magnifications.

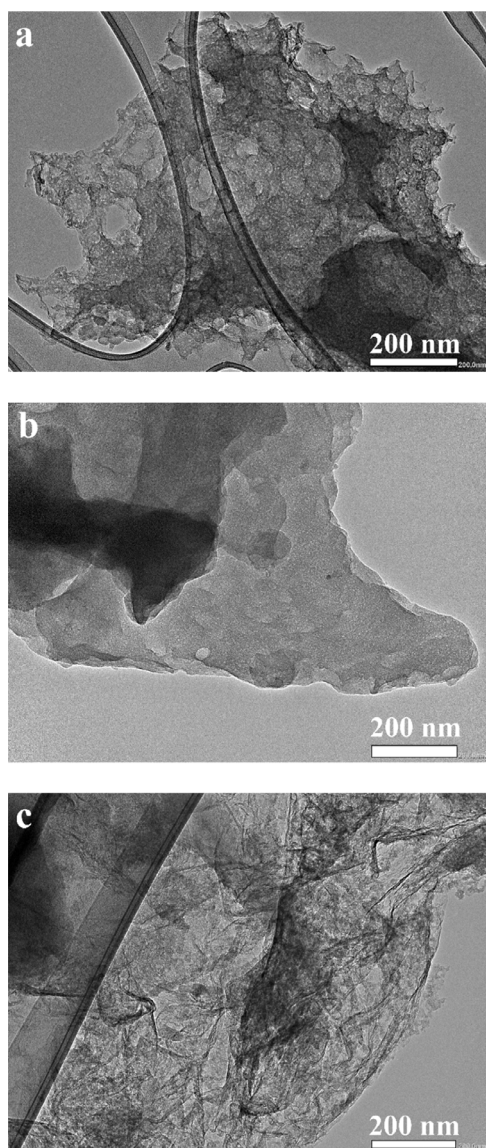


Fig. 5. Transmission electron micrographs of AC-1 (a), AC-2 (b) and AC-3 (c).

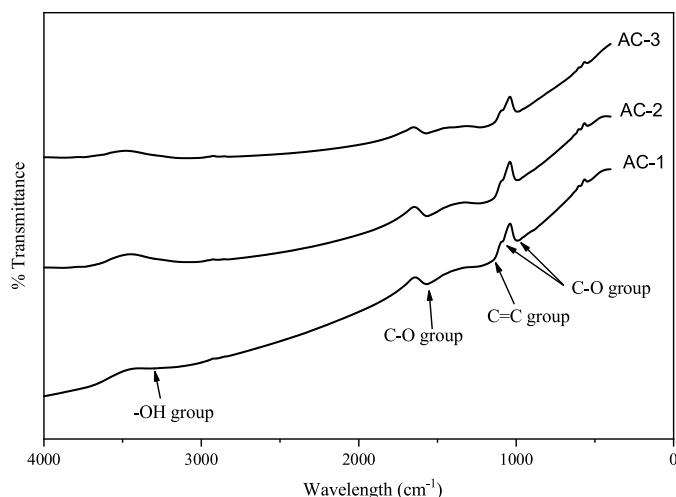


Fig. 6. FT-IR spectra obtained from AC-1, AC-2 and AC-3.

Based on proximate analysis, the hemicellulose, cellulose, lignin and ash content of the unprocessed CCP was 5.6, 28.3, 33.1% and 4.1%, respectively, which agreed well with literature values [8,9]. From elemental analysis, the C, H, N and O content of the raw CCP were 45.57, 5.25, 0.44 and 42.96%, respectively. Assuming the carbon content of hemicellulose, cellulose and lignin are 40, 44–50 and 55–65% [19–22], respectively, the carbon yield from CCP following carbonization was expected to be in the range of 33–38%. The carbon yield after carbonization was 26.3–28.5%, slightly lower than the expected value. However, some carbon loss during pyrolysis was to be expected.

The microstructural characteristics of unprocessed CCP are shown in Fig. S1 (supplementary material). Thin, folded sheets of organic material were observed and the surface appeared to be relatively smooth without pores.

3.2. Activated carbon characterization

3.2.1. Physical and chemical characterization

Following activation of the CCP-derived carbon, XRD patterns were obtained from AC-1, -2 and -3 as shown in Fig. 2. Two broad peaks were observed at $2\theta \approx 26.5^\circ$ and 44.5° assigned to the (002) and (100) planes of graphite (JCPDS no. 75-1621), respectively. No impurity phases were observed. The broadness of the peaks indicated low crystallinity of the graphite structure, which was expected in activated carbons. The large signal intensity in the low angle region, especially in AC-3, is an indication of high porosity within the samples [23].

The BET isotherms of all three activated carbon samples can be classified as Type I according to the IUPAC classification, indicating the predominance of micropores in all samples (Fig. S2, supplementary material). The N_2 adsorption volumes were similar in AC-1 and -2, while that of AC-3 was nearly four times higher. This was reflected in the significant improvement in surface area ($2056 \text{ m}^2 \text{ g}^{-1}$), total and micropore volume (1.17 and $1.06 \text{ cm}^3 \text{ g}^{-1}$, respectively) that was achieved in AC-3 (Table 1 and Fig. 3). The micropores mainly ranged between 1 and 2 nm in diameter, with the mode value for all three samples at 1.30 nm. Considering the solvated HSO_4^- ion size is 0.5–0.7 nm [24,25], the micropore sizes were sufficiently large to accommodate H_2SO_4 electrolyte ions.

The microstructure and morphology of the activated carbon samples observed by SEM (Fig. 4a–f) show an increased amount of porosity from AC-1 to AC-3. The pores appeared to be shallow in AC-1, while in AC-2 pores were connected and some pore-widening was observed. Macro- and mesopores, which are known to facilitate transport of electrolyte ions, were easily observed. In AC-3 (Fig. 4e–f), a rather dissimilar microstructure to AC-1 and -2 was observed, where the macropores appeared to be highly interconnected, while the surface of the carbon was rough and layered. TEM images (Fig. 5a–c) revealed the distinctions in agglomerate structure more clearly. In AC-1 and AC-2 (Fig. 5a and b, respectively) the pores were round, whereas the structure of AC-3 (Fig. 5c) appeared as wrinkled sheets. Domains of graphitic ordering were observed in all three samples though the structures were predominantly amorphous in agreement with XRD results.

The activated carbon samples were analyzed with FT-IR technique in order to determine the types of surface functional groups that were present. The vibrations were similar in all three samples, with five weak peaks located at ≈ 3300 , 1570, 1170, 1085 and 1000 cm^{-1} (Fig. 6). The peaks at 3300 and 1570 cm^{-1} correspond to $-\text{OH}$ and $\text{C}=\text{O}$ bond stretching in carboxyl groups [26]. The peaks at 1170 cm^{-1} and $1000\text{--}1085 \text{ cm}^{-1}$ are assigned to $\text{C}=\text{C}$ and $\text{C}-\text{O}$ stretching, respectively [26,27]. Oxygen in $\text{C}-\text{O}$ bonds could be present in a variety of surface functional groups. Acidic oxygen-containing surface functional groups include carboxyl, anhydrides, lactones and phenols. Basic oxygen functional groups include quinone, chromene and pyrone, while neutral groups are carbonyl and ether [28]. The acidic and basic groups may help with surface wettability of carbon, but may also cause a decrease in electronic conductivity. Oxygen surface functionalities could

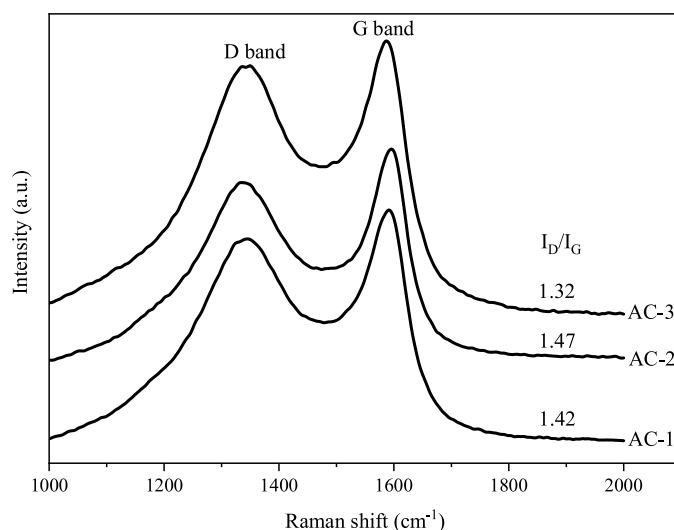


Fig. 7. Raman spectra obtained from AC-1, AC-2 and AC-3.

Table 2

Metal impurity content of AC-1, -2 and -3 determined by ICP-MS.

Sample	Amount (ppm)			
	Cr	Cu	Fe	Ni
AC-1	5.98	3.62	24.68	2.65
AC-2	9.87	5.97	15.34	6.02
AC-3	4.82	<0.01	2.48	<0.01

also enhance self-discharge when the carbon is used as supercapacitor electrode by catalyzing electrochemical oxidation or reduction of carbon and/or decomposition of electrolyte material [29,30]. As shown in our FT-IR results, the intensity of all observed peaks were low, indicating a minimal amount of oxygen surface functionalities. However, their presence was not unexpected as the activation temperature in this study was relatively low (700 °C).

Raman spectroscopy performed on AC-1, -2 and -3 all exhibited a D band at $\approx 1355 \text{ cm}^{-1}$ and G band at $\approx 1580 \text{ cm}^{-1}$ (Fig. 7). The G-band is attributed to the sp^2 graphitic structure, while the D band is attributed to defects in the carbon layers, namely, sp^3 -bonded (C–C or C–H) structure [26,31]. Also shown in Fig. 7 are values of I_D/I_G obtained from the integrated intensity ratio for each of the activated carbons. I_D/I_G is considered an indicator of the degree of graphitization within the

carbon [31]. Additionally, the value of I_D/I_G is inversely proportional to the average crystallite size in graphite materials [32]. I_D/I_G was lowest for AC-3, which indicates larger sp^2 domains were present. This could result in a slightly higher electronic conductivity for AC-3 compared to the other activated carbons. XPS analysis on AC-1, -2 and -3 C 1s spectra confirmed partial graphitization and the presence of oxygen surface functionalities and agree well with Raman spectroscopy and FT-IR results (Fig. S3, supplementary material).

Table 2 shows the metal content from the activated carbons measured by inductively coupled plasma-mass spectroscopy (ICP-MS). The major metals which affect the performance of supercapacitors are Cr, Fe and Ni as they can cause unwanted side reactions and reduce supercapacitor stability over time. The metal content of supercapacitor-grade activated carbons should be $<50 \text{ ppm}$ [33]. This high purity requirement is a major challenge in the production of these activated carbons and is one reason that supercapacitor-grade carbons are at least three times more expensive than commodity-grade activated carbon [34]. The ICP-MS results obtained from AC-1, -2 and -3 show that these carbons match the purity requirements for commercial supercapacitor-grade activated carbon.

3.2.2. Electrochemical characterization

The activated carbon samples were formed into electrodes and

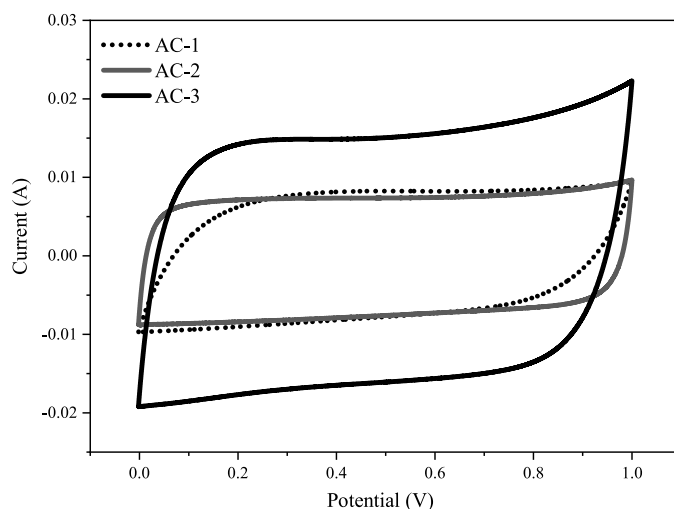


Fig. 8. Comparison of cyclic voltammograms obtained from AC-1, -2 and -3 at a potential scan rate of 10 mV s^{-1} .

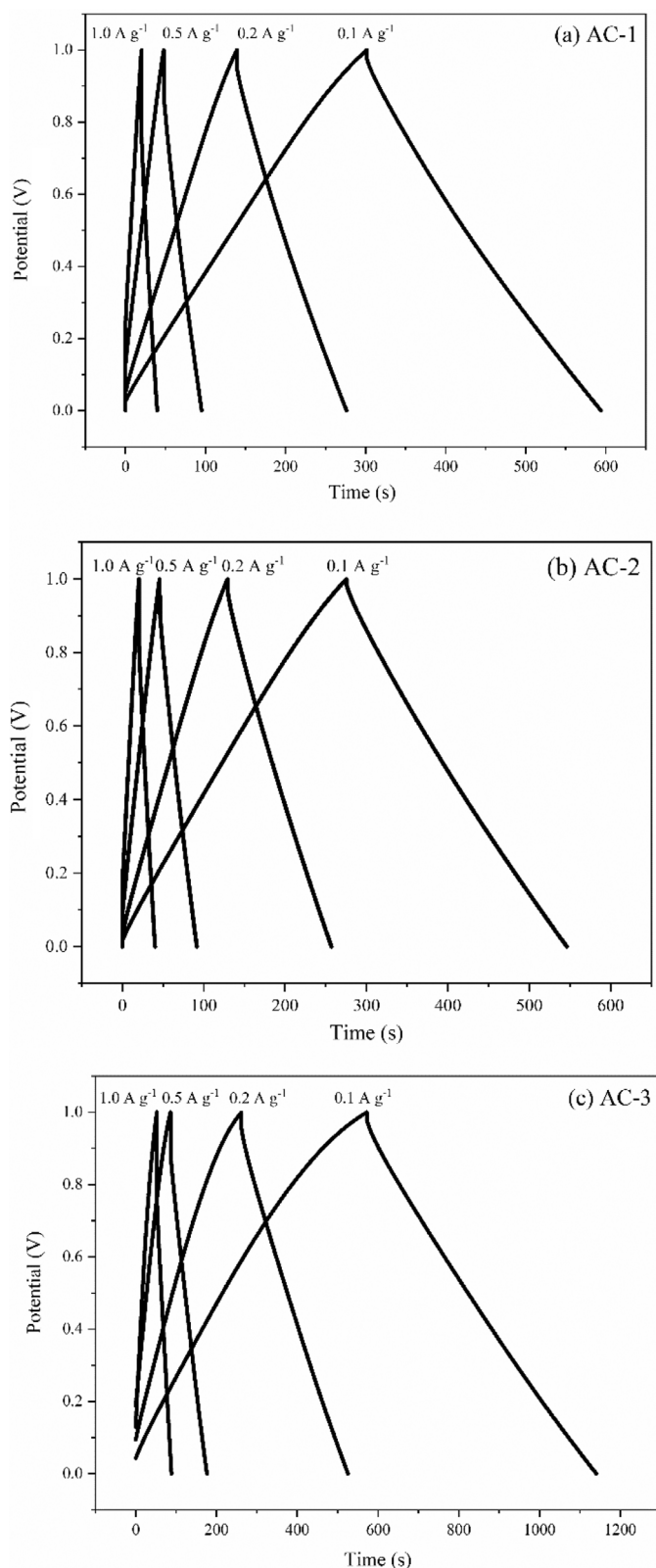


Fig. 9. Charge-discharge curves of (a) AC-1, (b) AC-2 and (c) AC-3 at different current densities.

tested for their electrochemical performance in symmetrical cells. A comparison of the cyclic voltammograms of AC-1, -2 and -3 at a potential scan rate of 10 mV s^{-1} is shown in Fig. 8. All activated carbons exhibited ideal rectangular-shaped curves without visible redox peaks.

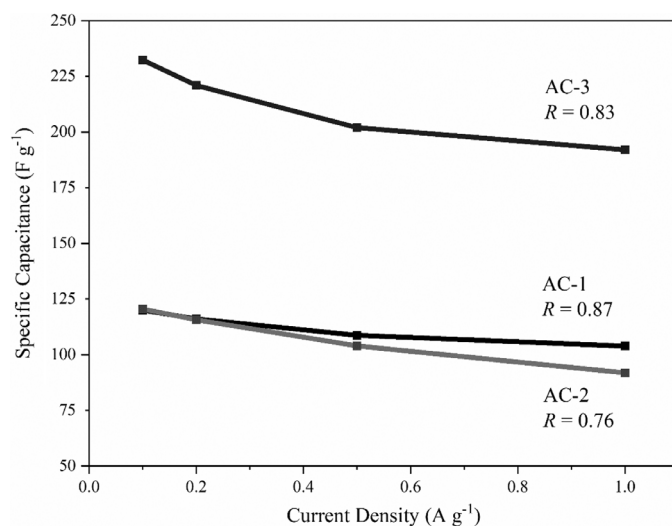


Fig. 10. Specific capacitance of AC-1, -2 and -3 measured at 0.1, 0.2, 0.5 and 1.0 A g^{-1} .

Table 3

Estimated C_{DL} and C_{PS} contributions to the total capacitance based on the BET surface area and experimental total capacitance values.

Carbon Sample	S_{BET} ($\text{m}^2 \text{ g}^{-1}$)	Estimated C_{DL} (F g^{-1})	Experimental C_{total} (F g^{-1})	Calculated C_{PC} (F g^{-1})	Calculated C_{PC} (%)
AC-1	633.2	63.3	120.01	56.7	47.2
AC-2	860.1	86.0	120.54	34.5	28.6
AC-3	2056.0	205.6	232.29	26.7	11.5

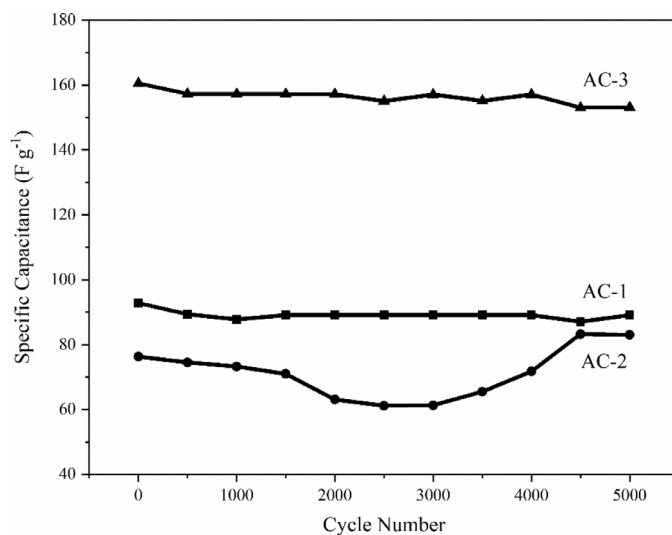


Fig. 11. Cycle performance of symmetrical AC-1, -2 and -3 pouch cells.

This does not preclude the presence of pseudocapacitance contribution to the capacitance as pseudocapacitance is better observed in three-electrode cells and is not always observable in CV results [35]. With increasing NaOH:C weight ratio, the CV curves became more rectangular and for AC-3, the plateau response current was twice as large as AC-1 and AC-2. Based on the physical and chemical characterization, it was expected that AC-3 would exhibit the best electrochemical performance amongst the three activated carbons, owing to its high micropore volume, higher graphitic content and its interconnected pore structure. CV curves were obtained from AC-3 using potential scan rates from 5 – 100 mV s^{-1} . Up to 20 mV s^{-1} , the curves retained a rectangular shape. At 50 and 100 mV s^{-1} the curves became less rectangular and

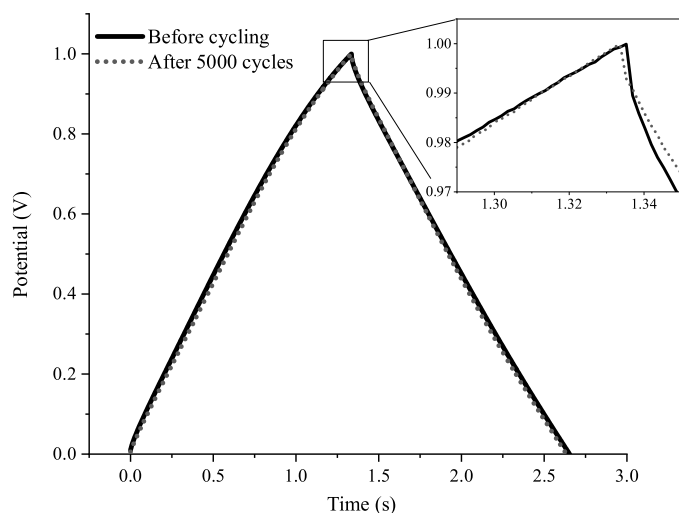


Fig. 12. Galvanostatic charge-discharge curve of AC-3 pouch cell before and after 5000 cycle tests at current rate of 0.5 A g^{-1} .

showed a tilt, especially at a scan rate of 100 mV s^{-1} , caused by the Ohmic resistance and slow ion diffusion kinetics (Fig. S4, supplementary material) [36].

The charge-discharge curves for AC-1, AC-2 and AC-3 are shown in Fig. 9. AC-1, -2 and -3 exhibited linear and symmetrical charge-discharge characteristics, a confirmation of the electric double layer charge-discharge mechanism. Pseudocapacitance may have been present at the lower current densities (0.1 and 0.2 A g^{-1}) but not visible in the galvanostatic charge-discharge profile. The IR drop values for AC-1 and -3 were similar, ranging between 3.28 – $4.33 \text{ } \Omega \text{ cm}^2$ for current densities between 0.1 and 1.0 A g^{-1} . The IR drop was slightly higher in AC-2 samples, 3.86 – $5.42 \text{ } \Omega \text{ cm}^2$. The specific capacitance values obtained from AC-1 to -3 at the measured current densities of 0.1 , 0.2 , 0.5 and 1.0 A g^{-1} are plotted in Fig. 10. In consistence with the CV data, the specific capacitance values of AC-3 were approximately twice as high as those obtained from AC-1 and -2. The specific capacitance obtained from AC-3 was 232.29 , 220.96 , 201.99 and 192.00 F g^{-1} at current densities of 0.1 , 0.2 , 0.5 and 1.0 A g^{-1} , respectively. These values are well within the range obtained from other biomass-derived activated carbons in literature (125 – 380 F g^{-1}) [36–40].

To further explain our results, the contribution of double layer capacitance (C_{DL}) to the total capacitance (C_{total}) was estimated using the BET surface area (S_{BET}) as shown in Eq. (2) [41].

$$C_{DL} = 0.10 \times S_{BET} \quad (2)$$

As the total capacitance is the sum of the double layer and pseudocapacitance (C_{PC}) contributions, the pseudocapacitance contribution was estimated using Eq. (3) [41]. The results of the estimations are shown in Table 3.

$$C_{PC} = C_{total} - C_{DL} \quad (3)$$

Based on our calculations, the pseudocapacitance contributions were not negligible. The pseudocapacitance originated from the oxygen surface functionalities as observed in the FT-IR and XPS results (Figs. 6 and S3, respectively). However, its contribution to total capacitance decreased with increasing NaOH:C ratio used during the activation process. In other words, with increasing micropore volume (and BET surface area), the micropore content contribution to the total capacitance became increasingly significant.

The rate capability (R) was determined by the ratio of the capacitance measured at 1.0 A g^{-1} to that at 0.1 A g^{-1} and shown in Fig. 10. It appeared that AC-1 had the best rate performance, but considering the small micropore content and modest capacitance value, the apparently high rate performance was likely due to the relatively smooth, pore-free surface as evidenced by the SEM images (Fig. 4a and b). On the other hand, AC-3 had both high capacitance and rate capability. Mesopores are known to contribute to high rate capability as they act as fast ion transport pathways. However, the pore analysis results showed that AC-3 had low mesopore content ($\approx 10 \text{ vol\%}$), thus the rate capability could be attributed to the interconnected macropore structure which acted as low tortuosity paths for quick ion access.

Based on EIS measurements, the charge transfer resistance of AC-3 ($0.46 \text{ } \Omega \text{ cm}^2$) < AC-2 ($1.85 \text{ } \Omega \text{ cm}^2$) < AC-1 ($3.33 \text{ } \Omega \text{ cm}^2$) (Fig. S5, supplementary material). Because there were no major differences in the type and amount of surface functional groups among the activated carbons, the reduction of charge transfer resistance with increasing NaOH:C ratio was attributed to the macropore structure, which allowed easy access of electrolyte to the carbon surface in different areas of the electrode. The slightly higher graphitic content of AC-3 compared to AC-1 and -2 as determined by Raman spectroscopy may have also assisted in charge transfer. The superior electrochemical properties of AC-3 compared to AC-1 and AC-2 point to the pertinence of the activated carbon surface area, micropore volume, micropore size and microstructure in accommodating electrolyte ions and allowing for efficient charge transfer which lead to higher capacitance. The interconnected pore structure of AC-3 may have had a large effect on the available surface area for electric double layer formation, compared to a closed pore structure, such as that observed in AC-1 and -2. There was clearly a

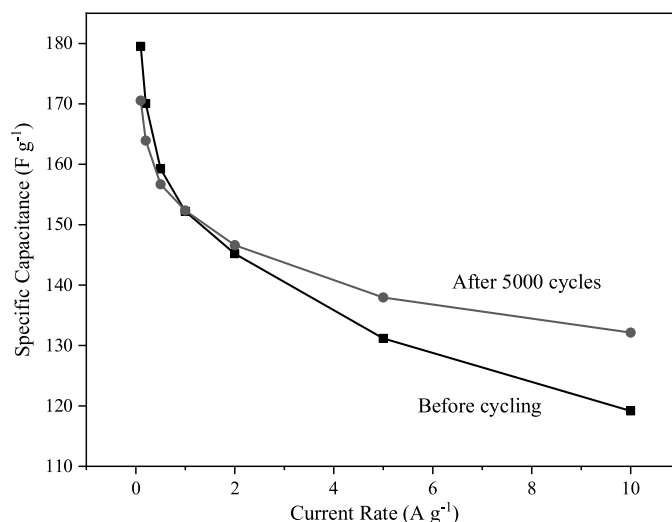


Fig. 13. Capacitance of a symmetrical AC-3 pouch cell at 0.1 – 10 A g^{-1} before and after cycling.

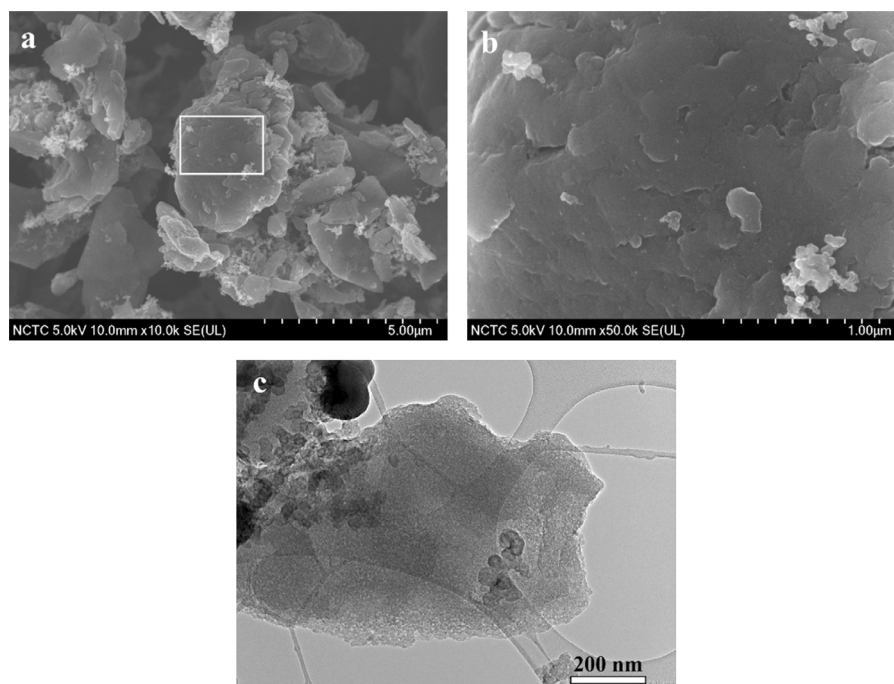


Fig. 14. Scanning electron micrographs at different magnifications (a, b) and transmission electron micrograph (c) of AC-3 composite electrode after operating at 0.5 A g^{-1} over 5000 cycles.

threshold level of activation at which this structure began to form, *i.e.*, where the carbon was sufficiently etched and the pores widened and connected, while maintaining structural integrity. In this study, this threshold was achieved using a ratio of $\text{NaOH:C} = 3:1$ by weight at an activation temperature of 700°C with CO_2 flow for 1 h.

To test the long-term cyclability of the performing activated carbons, pouch cells were cycled at 0.5 A g^{-1} over 5000 cycles. Pouch cells were used for long-term cycling as pellets tended to disintegrate over time. In Fig. 11, the capacitance values during cycling at a current rate of 0.5 A g^{-1} are shown for symmetrical pouch cells with AC-1, -2 and -3 electrodes. In agreement with the results from pellet electrodes, AC-3 pouch cells exhibited approximately twice the specific capacitance of AC-1 and AC-2 pouch cells, with AC-2 having the lowest capacitance. For the best-performing material, AC-3, the IR drop at 0.5 A g^{-1} was $1.04 \Omega \text{ cm}^2$ at the beginning of cycling tests and stabilized at $0.72\text{--}0.80 \Omega \text{ cm}^2$ after 2000 cycles, presumably due to improved electrolyte access to the pores with cycling. After 5000 cycles, approximately 95.4% of the initial capacitance was retained. The specific capacitance calculated from charge-discharge measurements at rates between 0.1 and 2.0 A g^{-1} after 5000 cycles remained similar compared to values prior to cycling. The charge-discharge curves before and after cycling showed little difference (Fig. 12). Interestingly, it appeared that the rate capability of AC-3 improved slightly after cycling, as can be seen by the increased capacitance at higher current rates ($5\text{--}10 \text{ A g}^{-1}$) (Fig. 13). The pre- and post-cycling R values were 0.85 and 0.89, respectively. The cause of the capacitance increase at higher current is not clear, though irreversible oxidation of the electrode upon cycling is one possibility. Despite our findings that the activated carbons in this study had only small amounts of oxygen functional groups, these groups could have facilitated the oxidation of the carbon surface.

Hydroxyl groups can be created which enhances wettability and improves specific capacitance [42]. The loss of carbon due to strong oxidation may have caused pore widening or facilitated the formation of mesopores. As summarized in Table 1, the pores in AC-3 were mainly micropores. Since macro- and mesopores are crucial as ion transport routes, the exposure to electrochemical cycling appeared to improve the rate capability by creating additional fast transport pathways or

widening existing ones [15]. Another possibility is that it may have taken a certain amount of cycling for electric double layers to stably form at the electrode/electrolyte interfaces due to the high surface area, which would explain the stabilization of IR drop after 2000 cycles [43]. Overall, little degradation in performance was observed in the AC-3 symmetrical cell when operated over 5000 cycles.

Post-cycling SEM and TEM micrographs of AC-3 composite electrodes are shown in Fig. 14. Those of AC-1 and AC-2 are included in Fig. S6 and S7, respectively. A mixture of activated carbon (micron-sized particles) and Super C65 conductive carbon was observed (smaller, sub-micron particles). The morphology and surface structure of AC-3 particles were similar to that observed prior to cycling (Fig. 4) but noticeably less pores were observed. The wrinkled sheet structure seen in TEM micrographs (Fig. 5c) was no longer detected. The loss of pores was not unexpected due to the possible blockage by the PVDF binder material and/or decomposition products of the electrode and electrolyte upon cycling [44]. This reduction in surface area along with the change in surface functionalities would eventually lead to the supercapacitor performance degradation upon further cycling.

In summary, activated carbon made from CCP has been shown to be a suitable candidate as an electrode material for supercapacitors due to its low cost, availability, relatively high specific capacitance, good rate capability and cycling stability.

4. Conclusions

Activated carbon was prepared from coconut coir pith using NaOH:C weight ratios of 1:1, 2:1 and 3:1 (AC-1, AC-2, AC-3). AC-3 yielded high surface area ($2056.0 \text{ m}^2 \text{ g}^{-1}$) and pore volume ($1.17 \text{ cm}^3 \text{ g}^{-1}$), with 91 vol% micropore content. Capacitance values of 232.3 F g^{-1} and 192.0 F g^{-1} were obtained at 0.1 A g^{-1} and 1.0 A g^{-1} , respectively. AC-3 capacitance retention was 95.4% after 5000 cycles at 0.5 A g^{-1} . The interconnected pore structure of AC-3 contributed to its performance by allowing full utilization of the surface area and efficient transport of electrolyte ions throughout the carbon structure.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.est.2019.100910](https://doi.org/10.1016/j.est.2019.100910).

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